Igniting Firm Dynamism through Skilled Migration: Evidence from Kazakhstan*

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Abstract

We study firm dynamics in a middle-income country following a high-skilled migration shock within an economic union. The shock ignites dynamism among mature small domestic firms, typically characterized as stagnant. We analyze Russian migration to Kazakhstan in 2022-23 using a shift-share strategy. More exposed districts experience 3.8 percentage points higher employment growth, more entry, and more exit. Lower-productivity districts experience higher firm turnover, while higher-productivity districts realize employment growth. Mature small firms contribute 24% of employment growth, despite driving most firm exit. Young small firms contribute 18% of employment growth without increased exit. Results suggest low-friction skilled migration fuels middle-income market dynamism.

Keywords: international migration, business dynamism, middle-income trap

JEL codes: F22, L25, D22, O12

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1 Introduction

This paper explores how a skilled migration shock can fuel firm dynamism in a middle-income economy. The question is timely given the scarcity of established remedies for business dynamism in these markets and limited evidence on how migration shapes the dynamics of firms over their life cycle. Middle-income economies are characterized by lower aggregate productivity (Bloom et al. 2010), stagnant firm lifecycles (Hsieh and Klenow 2014), and a distorted relationship between firm size and productivity (Restuccia 2025) that sustains a glut of unproductive, mature small firms (Li and Rama 2015). While literature attributes this stagnation to inferior management practices (Bloom et al. 2013; Bloom et al. 2020) or corruption and regulatory frictions (Restuccia 2025), evidence on specific solutions is thin. Skilled migration is a promising but under-scrutinized candidate; few studies decompose its effects on firm employment by both size and age. While Mahajan (2024) provides a benchmark U.S. analysis of unskilled migration flows, the consequences of a skilled migration shock for firms at different stages of their life cycle in a middle-income economy remain a critical open question.

The 2022-2023 migration of Russians into Kazakhstan presents such an unusual opportunity to assess firm dynamics in a middle-income economy. Kazakhstan provides data from public firm business registers that facilitate analysis of firm dynamics in response to a major shock. Kazakhstan matched Serbia as the top destination of Russians for the year 2022 with 150,000 migrants, who emigrated in response to international sanctions following Russia's invasion of Ukraine in late February and the war mobilization beginning September 21.[1] Low migration frictions and incentives for formal registration helped to facilitate cross-border labor movement and its observation. Kazakhstan and Russia are each among members of the Eurasian Economic Union (EAEU), which allows visa-free labor flows between the two countries. Immigration law motivates Russian migrants to enter formal employment, since a job qualifies them to obtain a temporary residence permit that allows a stay in Kazakhstan longer than 90 days. Digital government initiatives for Kazakhstan (EBRD 2022; World Bank 2024) have helped reduce frictions for migration, tax registration, formal employment, and new firm registration. Immediate access to

1. "Russians have emigrated in huge numbers since the war in Ukraine," 23 Aug 2023, *The Economist*, https://www.economist.com/graphic-detail/2023/08/23/russians-have-emigrated-in-huge-numbers-since-the-war-in-ukraine

the formal economy aligns with what Bahar et al. (forthcoming) highlight as the optimal condition for realizing the mutual economic benefits of a sudden migration shock for both migrants and the host country.

Unique geographic features facilitate the identification of the Russian shock's impact. Kazakhstan shares the world's longest continuous international land border with Russia and is less urbanized than similar middle-income nations (World Bank 2022), generating substantial variation across 215 districts in their appeal to migrants. To exploit this, we employ a shift-share design (Borusyak et al. 2025) that instruments for recent flows using the pre-shock spatial distribution of ethnic Russians. Common in the migration literature, this approach leverages well-documented network effects, where historical communities lower settlement costs for new arrivals (Altonji and Card 1991; Card 2001; Munshi 2003). We confirm the instrument's relevance through its strong correlation with both Russian permanent residence registrations and Russian-origin CV submissions. We support the validity of our design by documenting the absence of differential pre-trends in employment growth and ruling out key potential confounders, such as increased export demand and geographic exposure to Russia.

We estimate the impact of the migration shock using a difference-in-differences (DiD) framework on a district-level panel constructed from administrative firm data (2018–2023). Our primary specification interacts the pre-shock Russian population share in 2018 with a post-2022 indicator, including district and year fixed effects weighted by baseline district employment. The outcomes that we examine include employment growth, firm count growth, entry and exit rates, and output per worker. Given that the shock impacts both migrant and native labor through demand and supply channels, we focus on the aggregate equilibrium adjustments in exposed districts and how these vary by firm size, age, and ownership. Consistent with recent methodological guidance (Borusyak et al. 2025) and migration literature (Beerli et al. 2021; Abarcar and Theoharides 2024), we prioritize reduced-form estimates based on shock exposure, though we also report two-stage least squares (2SLS) results.

We find that districts more exposed to migration experience 3.8 percentage points higher employment growth and 5.0 percentage points higher firm count growth relative to less exposed areas. These aggregate effects are robust to discrete and continuous treatment

specifications and persist when excluding the two largest cities, Almaty and Astana. The more exposed districts also experience more firm entry, more firm exit, and higher output per worker. When we explore heterogeneity in effects based on pre-shock district-level labor productivity, we find evidence of productivity-based selection. The low productivity quartile experiences significant firm turnover, while the high productivity quartile drives firm expansion.

To further isolate mechanisms, we decompose employment growth by firm characteristics following approaches in Mahajan (2024). In contrast to high-income economies, small mature firms (age > 5, size < 50) drive a substantial 24% of employment growth. Crucially, these small mature firms exhibit high turnover. They account for the majority of firm exits—a volume equivalent to 40% of the total net firm count increase. Conversely, small young firms exhibit no increased exit, contributing to growth purely through expansion and new entry (each adding $\sim$18%). While both domestic and foreign entities expand, the response is driven by domestic ownership, with domestic small and medium firms accounting for roughly three-quarters of the total employment increase.

This is the first study to analyze firm dynamics by both size and age following a migration shock in a middle-income economy. While a growing literature links migration to firm-level employment, evidence on how these effects vary by firm size remains scarce. Moreover, existing studies focus overwhelmingly on high-income economies, including the U.S. (Olney 2013; Waugh 2017; Orrenius et al. 2020; Mahajan 2024), Germany (Dustmann and Glitz 2015), France (Mitaritonna et al. 2017), and Switzerland (Beerli et al. 2021). Among the few studies in developing or transition settings, the focus remains on aggregate outcomes rather than heterogeneous firm adjustments. Aksoy et al. (2025) analyze Ukrainian migration to Poland—a recently graduated high-income economy—but restrict their scope to new firm registrations. Similarly, research on Syrian refugees in Turkey examines general entry, exit, and employment (Aksu et al. 2022) or industry concentration (Akgündüz et al. 2023). Decomposing these results by firm size and age allows us to isolate specific margins of adjustment driving firm dynamism.

We contribute to the broader migration literature by providing a firm-level parallel to the labor market finding that immigrants complement rather than substitute for natives (Peri and Sparber 2009). In the realm of entrepreneurship, a central question is whether

migrant business creation "crowds out" native activity (Fairlie and Meyer 2003; Fairlie and Lofstrom 2015; Ghimire 2021) or encourages it (Wallskog 2025). We find no evidence of displacement among young firms; instead, exit is concentrated entirely among mature small firms. This pattern aligns with allocative efficiency rather than harmful crowd-out, as the employment loss from mature exits amounts to less than one-third of the net employment expansion arising through new and young small firms.

To the firm dynamics literature, we contribute unique evidence that small mature firms can make substantial contributions to employment growth in middle-income countries, contrasting with U.S. evidence. Foundational studies for the U.S. establish that small firms are not drivers of net employment growth (Davis et al. 1996) and that smallness has no independent correlation with growth once age is controlled for (Haltiwanger et al. 2013)—results further confirmed using a shift-share design based on manufacturing demand shocks (Adelino et al. 2017). In contrast, descriptive analyses and calibrations for middle-income countries suggest stunted growth among medium and large firms, with mature small firms joining young small firms as primary drivers of employment growth (Bartelsman et al. 2009; Hsieh and Olken 2014; Li and Rama 2015; Rijkers et al. 2017). By leveraging the Russian migration shock to Kazakhstan, we provide the first evidence with a causal research design confirming these distinct structural features of a middle-income economy. This is also among the first studies using causal methods to examine how a migration shock reshapes firm-level dynamics in a post-Soviet economy, where data availability has historically constrained such analysis.

Our paper proceeds as follows. Section 2 provides relevant background on EAEU migration policy and other accounts of the Russian emigration. Section 3 summarizes our data sources for Kazakhstan and provides descriptive analysis confirming the regional dispersion of Russian immigration to Kazakhstan. Section 4 discusses our methods and research design, Section 5 presents our main regression results, and Section 6 concludes.

2 Background

We provide here additional sources on the context in which the Russian immigration to Kazakhstan is occurring. First, we discuss migration policies of the Eurasian Economic

Union (EAEU) and Kazakhstan's ability to facilitate migration. We then summarize existing evidence on the composition of the Russian immigration and whether the Russians are mere transient migrants.

2.1 The Eurasian Economic Union

Russian citizens can relocate to Kazakhstan with minimal frictions thanks to the Eurasian Economic Union, which was established in 2015 with the aim of creating a common market with free movement of goods and services. Kazakhstan and Russia are both members, along with Belarus, Kyrgyzstan, and Armenia. The principle of free flow of workforce within the Union allows Russian citizens to engage in labor activities in Kazakhstan without obtaining a work permit. All that member citizens require is a temporary residence permit, which is issued by registering a local Kazakhstani labor contract. Without any residence permit, Russians can still stay 90 of the previous 180 days in Kazakhstan. The geographic proximity of EAEU members as well as common knowledge of the Russian language among members are additional factors facilitating immigration between members.

EAEU immigration into Kazakhstan is also facilitated by Kazakhstan's general digital transformation initiatives. Processing for obtaining an individual identification number (IIN), necessary for many economic activities, is practically instant. For starting a firm, the typical processing time is less than a month.

2.2 Skill composition of Russian migrants

Both academic studies and journalistic sources support that the Russian immigrants are largely skilled. The most rigorous study of the skill composition of Russian immigrants, Wachs (2023), suggests large flows of ICT workers. The study, based on GitHub developer profiles, finds that Georgia and Armenia are preferred destinations for ICT workers, but a large flow into Kazakhstan is still observed. The results of the study are consistent with the journalistic consensus that the Russian migration is skill abundant. For example, a BBC article from March 2022 just after the invasion emphasizes

Russia's brain drain.[2] A *Washington Post* article from May 2022 emphasizes the devastation to Russia's information technology industry.[3] While ICT workers represent a well-documented component of the outflow, the Russian emigration more broadly drew from educated, professionally active segments of the population, consistent with the brain drain characterization in contemporary accounts.

2.3 Duration of Russian migration

Available data sources confirm that the Russian immigration we observe is not purely transitory, or just Russians transiting through Kazakhstan. The 150,000 estimate for the Russian immigrants into Kazakhstan provided in *The Economist* (fn 1) is ultimately sourced from Kazakhstan's Ministry of Internal Affairs, who additionally assert that these 150,000 Russians remained in Kazakhstan at the end of 2022.[4] Survey data on Russian emigrants from the OutRush project reveal that their initial plans in March 2022 were for short-term stays but they eventually recognized the need for a more permanent stay in their destination countries.[5]

Even if Russians eventually return, empirical evidence from other immigration waves suggests that Russians may still maintain mutually beneficial trade links with Kazakhstan. Such was the case for early 1990s Yugoslavia refugees who immigrated to Germany, and later returned home, but still maintained trade ties with Germany (Bahar et al. 2022).

3 Data and descriptive analysis

This section introduces the data sources and presents descriptive statistics on firm dynamics and migration inflows. The data combine rich firm-level microdata with district-level migration indicators. The descriptive patterns point to marked differences in firm

2. Rayhan Demytrie, "Russia faces brain drain as thousands flee abroad" 13 March 2022, BBC, https://www.bbc.com/news/world-europe-60697763

3. Anthony Faiola, "Mass flight of tech workers turns Russian IT into another casualty of war," 1 May 2022, *Washington Post*, https://www.washingtonpost.com/world/2022/05/01/russia-tech-exodus-ukraine-war/

4. "Escape from War: New data puts the number of Russians who have left at more than 800,000 people," 28 July 2023, *RE:RUSSIA*,https://re-russia.net/en/review/347/

5. Nina Kostenko, Margarita Zavadskaya, Emil Kamalov, and Ivetta Sergeeva, "The Russian Rhizome: A Social Portrait of New Immigration," 11 Jan 2023, *RE:RUSSIA*, https://re-russia.net/en/expertise/045/

growth by size and age as well as heterogeneous migration inflows, providing the foundation for the empirical analysis.

3.1 Data sources and summary statistics

Our analysis combines two complementary microdata sources from Kazakhstan's Bureau of National Statistics (BNS). For firms with fewer than 100 employees, we rely on the annual public Business Registers Data (BRD) for 2018–2023, which reports legal entities by employment band, ownership structure, type of economic activity, region, and registration date. While this dataset does not provide firm-level output or balance sheet information and reports employment only in size bands, the bins for small firms are relatively fine (1–5, 6–10, 11–15, 16–20, 21–30, 31–40, 41–50, 51–100 workers), allowing us to distinguish employment dynamics among smaller firms with sufficient granularity. In our analysis, we use only BRD bins up to 100 employees, since larger firms (with 100 or more employees) are covered by the confidential firm-level dataset from the BNS. We combine the two sources to ensure coverage of the full firm size distribution: the confidential data provide precise employment counts for larger firms, while the public BRD captures the universe of smaller firms. In our baseline, we assign the minimum value of each size band to small firms and use exact confidential values for larger firms. As a robustness check, we also re-estimate results using the midpoint and maximum of each size band for smaller firms.

We also draw on district-level aggregates from the BNS to construct outcomes beyond firm-level employment. Log output per worker is derived from BNS regional national accounts, which report annual value added at the district level at current prices; we deflate using the national CPI to obtain real values. Firm count growth is constructed directly from the business register data described above.

In addition to the firm-level data, we draw on several datasets capturing migration-related indicators. These include annual BNS data on the share of the population of Russian nationality for each of Kazakhstan's 215 districts, annual BNS data on the number of residence permits issued to Russian nationals, and yearly counts of individual identification numbers (IINs) assigned to foreign nationals from the Committee of the Migration Service of the Ministry of Internal Affairs. While the IIN data are not used

directly in our regression analysis, they provide a useful administrative benchmark in the summary statistics, since obtaining an IIN is a prerequisite for residence, employment, and access to public services. We also use yearly counts of newly posted CVs from HeadHunter.kz, the largest online job-search platform in Kazakhstan, as a proxy for job search activity and labor market integration. Because migration inflows and economic activity are heavily concentrated in the two largest cities, Astana and Almaty, we report results both including and excluding them throughout the analysis.

Table 1 reports summary statistics for the key variables used in our analysis. In 2018, the average district in our largest sample (including Astana and Almaty) had a Russian population share of 20 percent, with substantial variation across districts. Average total employment per district is 22,915 workers. Between 2018 and 2023, districts issued an average of 24 permanent residence permits to Russian citizens per year, although the distribution is highly skewed. The average year-on-year change in the log number of permits is 0.055. Additional variables include log output per worker, employment growth, and firm count growth rates at the district level, reported as averages over the 2018–2023 period.

While the aggregate Russian inflow to Kazakhstan after 2022 far exceeded the number of new permanent residence permits, these permits provide the only district-level administrative measure available for the full period. Data on temporary permits or IIN registrations exist only at the national level and cannot be linked to districts. We therefore interpret permanent residence permits as capturing the subset of migrants who have settled and integrated locally—those most likely to affect employment and local demand. This interpretation aligns with international evidence showing that obtaining a residence or work permit is closely tied to migrants' formal labor market participation (Buechel et al. 2023; Oğuş Binatlı and Esen 2017). Similarly, the OECD International Migration Outlook treats newly issued residence and work permits as standard indicators of migration inflows across countries (OECD 2024). Finally, validation studies by the World Bank and LinkedIn show that changes in online job-search activity track official migration patterns at high frequency, confirming that CV-based digital data provide credible proxies for real migration flows (World Bank and LinkedIn 2018).

[Table 1 here]

Table 2 summarizes summary statistics by firm size–age category. Panel A reports aggregate employment growth, our main dependent variable, by year and category. Firms are classified as small–young (1–50 employees, age 1–5 years), small–old (1–50 employees, age over 5 years), medium (51–250 employees), and large (over 250 employees). Firm age is calculated as the difference between the observation year and the registration year, with the five-year cutoff for "young" firms following the U.S. Census Business Dynamics statistics. Age data are unavailable for medium and large firms because registration year is not reported in the confidential dataset, so these categories are not split by age. Employment growth is measured using the average of current- and previous-year employment as the denominator (see Section 4.1). In 2022 and 2023, employment growth was negative for small–young and large firms, and positive for small–old and medium firms, though in 2023 the positive growth was lower than in 2022. Panel B presents the share of firms by category and year, showing that most firms are small, with a roughly equal distribution between small–young and small–old. Between 2022 and 2023, the share of small–young firms declined, while the share of small–old firms increased. Panel C reports the distribution of total employment by size–age category, highlighting that medium and large firms together account for over 60% of aggregate employment throughout the sample period. While large and medium firms employ the majority of workers, the employment share of small–old firms has gradually increased over time, whereas the share of small–young firms rose in 2022 before slightly declining in 2023.

[Table 2 here]

3.2 Migration indicators and instruments

The scale and timing of Russian immigration are clearly visible in national administrative and job-search data, which provide high-frequency measures of inflows and labor market engagement. While our firm-level microdata do not record individual arrivals, national administrative records track key steps such as obtaining individual identification numbers (IINs) and residence permits, and online job-search platforms record CV postings by Russian citizens. These indicators offer a clear timeline of the migration surge, highlighting its main spikes and showing whether elevated levels persisted beyond the initial shock.

Figure 1 illustrates two national indicators of Russian migration to Kazakhstan. Panel A shows a dramatic surge in the issuance of IINs to Russian citizens: in 2022 the number issued was more than 20 times higher than in 2021, reflecting the large migration inflows following the invasion of Ukraine and the partial mobilization. While the total declined in 2023, it remained over 14 times the 2021 level, indicating a sustained elevated inflow relative to pre-war conditions. Panel B shows a parallel trend in labor market participation, with the number of CVs posted by Russian citizens on HeadHunter.kz roughly doubling between 2021 and 2022 and increasing by a further 40 percent in 2023. Together, these series show both a sharp rise in migration and sustained high levels of job-search activity in the post-war period.

[Figure 1 here]

The distribution of Russian population across Kazakhstan is important to describe, because it relates to actual migration flows and plays a lead role in our identification strategy. As shown in Figure 2, this distribution is far from uniform: Russian population shares are heavily concentrated in the northern and northeastern districts bordering Russia, with much lower shares in the south and west. This spatial variation underlies our main empirical measure of exposure to migration, which uses the continuous 2018 Russian population share. Additionally, we use a binary measure of exposure, based on whether a district's 2018 Russian population share is above or below the national median among districts.

[Figure 2 here]

Figure 3 offers descriptive evidence that the Russian population share relates to actual migration flows, a key premise for our research design. We plot and compare residence permits and CV postings across above- and below-median districts by Russian population share. Panel A shows that the number of permanent residence permits issued to Russian citizens rose sharply after 2021 in both groups, with higher levels and substantially larger percentage increases in the above-median group, rising by roughly two-thirds from 2021 to 2022 and doubling from 2022 to 2023 compared to increases of about 25% and 50%, respectively, in the below-median group. Panel B shows a similar pattern for CVs posted

by Russian citizens on HeadHunter.kz, reflecting active labor market participation, with a more pronounced post-2021 increase in the above-median group. To further support these relationships, Appendix A.1 develops event study estimates capturing how each migration inflow measure relates to the continuous Russian population share. The results align well with our descriptive plots: there are no pre-trends, while we estimate statistically significant coefficients for the effect of Russian population share on each migration indicator for 2022 and for 2023.

[Figure 3 here]

4 Empirical framework

4.1 Difference-in-differences approach

We first present the variables and equations that we will use for our difference-in-differences (DiD) estimation. We present these first to clarify our use of common DiD approaches, before we provide detailed justification for our strategy in the next subsection.

Our main dependent variables include employment growth, firm count growth, firm entry and exit rates, and log output per worker, all measured at the district level. For the growth-rate outcomes, we adapt the definition of Davis et al. (1996) widely used in the business dynamics literature. For each measure, we first aggregate the relevant quantity (total employment or total number of firms) across all firms within a district for each year, and then compute growth as the change relative to the average of the current and previous period: $gr_{dt} = \frac{n_{dt}-n_{dt-1}}{(n_{dt}+n_{dt-1})/2}$, where n_{dt} denotes either total employment or total firm count in district d and year t. This symmetric measure reduces distortions from using only the previous period as the base, remains bounded, and allows consistent interpretation across districts and years.

We use an employment-weighted difference-in-differences (DiD) estimator to evaluate the net employment effect of Russian immigration, and we ultimately provide standard checks necessary for DiD validity. Our estimation uses employment weights so that coefficients can be interpreted as their effect on aggregate employment, following, e.g., Haltiwanger et al. (2013), whereas equal-weighting would give outsize influence in results for the outcomes of micro or small firms.

The time variable for our DiD is War_t, a binary indicator that is defined as 0 during the pre-war period and 1 following the start of the war in 2022. So we consider one change in 2022 and avoid well-documented concerns related to staggered DiD (Roth et al. 2023).

Our variable capturing the variation in intensity of treatment is $Share_d$, representing the 2018 ethnic Russian population share for each of 215 districts in our main specifications, or 204 excluding Astana and Almaty. We anticipate this variable to be a relevant predictor of Russian immigration and unrelated to the dependent variable, after conditioning on district fixed effects and year fixed effects. The distribution of $Share_d$ is shown in Figure 2.

We estimate the following baseline two-way fixed effects equation:

$$y_{dt} = \alpha + \delta(\text{Share}_d \times \text{War}_t) + \omega_d + \mu_t + \epsilon_{dt} \tag{1}$$

where y_{dt} denotes the district-level outcome variable, and Share_d is the 2018 ethnic Russian population share for district d. The main coefficient of interest is δ on the interaction term $\text{Share}_d \times \text{War}_t$, capturing the effect of the Russian immigration shock on district-level outcomes. Equation (1) is estimated separately for each dependent variable: employment growth and firm count growth (each constructed as gr_{dt} above), firm entry and exit rates, and log output per worker. We use ω_d and μ_t as fixed effects representing the district and year fixed effects, respectively. For standard errors and inference, we cluster by district, because the district is the dimension along which our instrument for treatment varies (Abadie et al. 2022). Clustering addresses any serial correlation within districts. We always have over 200 districts, a sufficient number for clustering asymptotics. The error term, ϵ_{dt}, is otherwise standard.

The DiD approach naturally relies on a form of parallel trends assumption, which implies an absence of any pre-trends in employment growth. So we estimate a standard event study specification which uses the year before treatment 2021 as a baseline and then includes additional interaction coefficients for 2019 and 2020, each year prior to treatment. We expect to find no effects for either 2019 or 2020 and no trend in point estimates. We estimate these using a sup test and roughly uniform confidence intervals following the advice of Freyaldenhoven et al. (2019) and code by Rios-Avila (2023).

Our regression analysis is conducted for both employment growth and firm count

growth using a parallel set of specifications. For each outcome, we use the continuous 2018 Russian population share as our primary migration exposure measure and also consider a binary version of this variable, equal to one for districts above the median share. For both the continuous and binary measures, we present estimates including and excluding Kazakhstan's two major cities, Almaty and Astana. Excluding the major cities confirms that we are truly deriving results from regionally-diffuse migration.

4.2 Research design

We categorize discussion here according to the relevance of the treatment variable, exclusion restrictions, stable unit value treatment assumption (SUTVA), anticipation, and heterogeneity of treatment effects.

Our research design relies on the relevance of the Russian native population share in determining the intensity of impact of the shock of the war. A large literature reflects that network effects and existing diasporas are relevant in determining migration location decisions (Altonji and Card 1991; Card 2001; Munshi 2003; Jayet et al. 2010; McKenzie and Rapoport 2010; Theoharides 2018). The Russian population shares also reflect optimal migration decisions before the war, and these decisions would likely be based on some persistent factors like geography, whose consequences were then amplified in 2022 by many more Russians making these same decisions. This attractiveness could also lead to violations of the parallel-trends assumption, but not necessarily so, since other economic factors could still result in similar pre-war employment growth rates across districts that are attractive to Russians. As a robustness check, we provide results confirming that our DiD approach captures an increase in permanent residence permits and CV postings with higher ethnic Russian populations, so our measure of treatment intensity does relate to a headline measure of immigration.

Our research design also requires a standard exclusion restriction, that there is no causal pathway from the 2018 Russian population share to employment growth other than through Russian immigration, after conditioning on district unobservables common to all years and year unobservables common to all districts. District fixed effects close down a number of potential violations involving district characteristics that are constant over time. Potential violations in periods other than 2022 could be revealed from the pre-

trends analysis. The only potential remaining concern then is region-specific (i.e., shocks at the broader administrative level grouping multiple districts) and 2022-specific shocks related to Russian share that are not directly linked to immigration. The major cities Almaty and Astana have relatively large Russian shares and may face year-specific urban-specific shocks, but we address this concern by focusing more on estimations excluding these cities. Otherwise a violation of the exclusion restriction is difficult to identify.

Regarding the stable unit treatment value assumption (SUTVA), SUTVA is always difficult to justify fully in spatial data given the possibility of spillovers. We do not model spillovers explicitly, and our baseline specification includes district and year fixed effects but no neighbor-exposure terms. The exclusion of Almaty and Astana in some specifications reduces concerns that results are driven by dynamics concentrated in the largest urban centers, yet it does not rule out spillovers across adjacent districts elsewhere. Consequently, our estimates using the continuous exposure measure ($Share_d$) should be interpreted as reduced-form intent-to-treat effects that may combine direct impacts within a district with short-range spillovers to neighboring areas. In this sense, the identifying assumption is not strict SUTVA but a weaker form that allows for limited local interference.

To examine potential nonlinearities in the response to exposure, we also estimate a binary specification in which the treatment indicator equals one if a district's pre-war Russian population share is above the median ($above\text{-}median_d$). This alternative captures the idea that additional Russian immigration may occur only once a critical mass of pre-existing Russians is reached, rather than varying proportionally with initial shares.

Regarding the absence of anticipation, we can check pre-trends to address the possibility of migration in advance of the war. The administrative and job-search data in Figure 1 show no meaningful increase in the issuance of individual identification numbers or in CV postings by Russian citizens prior to 2022, indicating that large-scale migration flows did not occur in anticipation of the invasion.

Heterogeneity of treatment effects in a continuous DiD design has only recently received attention, and evaluating potential related concerns in our setting is necessary. Our design relies on comparing the effect of presumably large Russian immigration flows to a district with a large incumbent Russian populations, against the effect of presumably

smaller Russian immigration flows to a district with a smaller Russian population. Ideally, however, the counterfactual for a high-Russian-share district would be its own outcome under a smaller inflow. If districts with larger Russian populations are better able to integrate a given inflow, the estimated difference between high- and low-share districts could overstate the true causal effect, as discussed in Callaway et al. (2021). In this case, the effect that we estimate is likely larger than the ideal counterfactual, so identifying a causal parameter in our setting requires assumptions beyond standard parallel trends. What is helpful in our setting is that the map in Figure 2 and descriptive analysis suggest concentration of Russians in certain districts, and the potential bias is likely small to the extent that regions with large Russian populations share similar integration capacities, while regions with low Russian populations attract few immigrants and thus provide a plausible counterfactual for a near-zero immigration shock. This reasoning also motivates estimating an alternative specification that replaces the continuous Russian population share with a binary indicator for above- versus below-median share. Both specifications are presented in our main results tables. The binary specification also provides a partial check on the concern raised by Callaway et al. (2021): because it compares districts discretely classified as above- versus below-median rather than relying on proportionality between Russian population share and inflows, it is less vulnerable to the bias that would arise if high-share districts are systematically better at integrating migrants.

4.3 Decomposition of growth

Finally, we decompose the estimated post-war effects on employment and firm count growth following Mahajan (2024) to identify which types of firms, by size, age, and ownership, drive the aggregate employment and entry–exit dynamics observed in the baseline results.

We partition firms into four size–age categories $g \in \{\text{small–young}, \text{small–old}, \text{medium}, \text{large}\}$, as defined in Section 3. Within each category, we express the growth rate as:

$$gr_{dgt} = \sum_{j \in \mathcal{E}_{dgt}} Emp_{j,t+1} - \sum_{j \in \mathcal{X}_{dgt}} Emp_{j,t_0} + \sum_{j \in \mathcal{C}_{dgt}} \Delta Emp_j, \tag{2}$$

where $\mathcal{E}_{dgt}$, $\mathcal{X}_{dgt}$, and $\mathcal{C}_{dgt}$ denote, respectively, the sets of entering, exiting, and continuers

in district d, category g, and period t. Dividing the right-hand side by average employment in the same cell gives the relative contributions of entry, reduced exit, and continuers to total employment growth. For firm count growth, the same logic applies but the continuer term drops out, since each firm contributes one unit to the total. Hence, firm count growth is decomposed only into entry and (reduced) exit components.

The decomposition is normalized within each breakdown across size–age groups: (i) in the *Total* column, the four coefficients across g sum to one; (ii) in the *By firm dynamics* panel, coefficients across all size–age groups and components sum to one, and, by construction, the three components for a given group sum to that group's total share; (iii) in the *By ownership* panel, coefficients across all size–age groups and ownership types sum to one, and the domestic and foreign shares for a given group sum to that group's total share.

Thus, coefficients can be read as shares of *total* post-war growth attributable to each size–age group and channel. The exit component is entered with the opposite sign so that reductions in exit appear as positive contributions; entry and exit together match the group's total share.

5 Results

5.1 Baseline results for firm growth and turnover

The first step of our analysis is to confirm, in Figure 4, the absence of pre-trends that would invalidate the parallel trends assumption of our DiD design. Panel A plots estimates using the continuous 2018 Russian population share, while Panel B uses a binary indicator for above- versus below-median Russian share. In both cases, the figure evaluates pre-trends for our baseline specification at the district level, including Astana and Almaty. We estimate relationships for the interaction of the Russian population measure and each year 2019, 2020, 2022, and 2023 relative to 2021, including district and year fixed effects. We fail to reject an effect of zero for 2019 and 2020, and there is no discernible trend in point estimates during the pre-war years (2019–2020). This supports the interpretation that the increase in employment growth from 2021 to 2023 does not reflect a continuation of prior trends of districts with higher Russian populations experiencing increases in their

employment growth rates.

[Figure 4 here]

Table 3 presents our baseline estimates of the effect of Russian migration on district-level employment growth. Across all four specifications, the interaction between the 2018 Russian population share and the post-war period (2022–2023) is positive and statistically significant at conventional levels. In column (1), the coefficient of 0.064 implies that in a district with the mean Russian population share (20 percent), employment growth was about 1.3 percentage points higher after the war (0.064×0.20) relative to districts with no Russian population, holding other factors constant. Column (2) replaces the continuous share with a binary indicator for above-median Russian population share. The estimated effect of 0.038 suggests that high-share districts experienced employment growth 3.8 percentage points higher than low-share districts after the war. This somewhat larger magnitude likely reflects the discrete gap in exposure between the two groups and a mild non-linearity in the response, whereby employment gains increase once the local Russian population surpasses a critical mass.

[Table 3 here]

Columns (3) and (4) exclude Kazakhstan's two largest cities, Almaty and Astana, to ensure results are not driven by dynamics in major urban centers. The estimated effects remain positive and significant, though slightly smaller in magnitude, indicating that the relationship between Russian population share and post-war employment growth is not solely an urban phenomenon.

Next, we explore the mechanisms driving employment growth, starting with growth in the number of firms in Table 4. In column 1, using the continuous 2018 Russian population share, we estimate a statistically significant coefficient of 0.099 ($p < 0.01$). Evaluated at the mean district Russian population share of 20 percent, this implies that, in 2022 and 2023, districts with the average Russian share experienced firm count growth rates about 2.0 percentage points higher than otherwise similar districts. Column 2 shows that the results are robust when using a binary indicator for above- versus below-median Russian share. Columns 3 and 4 report the same specifications excluding Astana and Almaty; the

estimated effects remain positive and statistically significant, though somewhat smaller in magnitude.

[Table 4 here]

In Table 5 and Figure 5, we find that employment growth is accompanied by higher firm turnover and output per worker—outcomes characteristic of business dynamism. More-exposed districts experienced a statistically significant increase in both entry and exit rates. Output per worker also rises significantly in exposed districts, suggesting that the shock-induced restructuring was accompanied by improvements in aggregate performance. The figure's event-study estimates confirm that pre-trends remain absent, while firm entry and exit exhibit staggered timing consistent with canonical theories of competition and industrial dynamics. Firm entry increases contemporaneously with the migration shock in 2022, consistent with immediate increases in market demand and entrepreneur supply. Firm exits occur later in 2023, indicating a lagged response to increased competition. Effects on district output per worker are evenly distributed across time. Taken together, the joint pattern of higher firm turnover and higher output per worker is consistent with productive reallocation: resources shifting away from less viable firms toward entrants and surviving firms better suited to the post-2022 environment.

[Table 5 and Figure 5 here]

For further confirmation that firm turnover and expansion are driven by productivity-based selection, we estimate effect heterogeneity based on pre-shock district labor productivity quartiles. Using the interquartile range (p25–p75) as the reference category, we focus on effects in the lowest quartile (p0–p25) and highest quartile (p75-p100). Results are consistent with productive reallocation. Districts with low productivity experience significantly stronger increases in both entry and exit. High-productivity districts experience greater employment growth. The patterns together are consistent with a reallocation channel in which high-productivity firms expand in response to positive labor demand and supply shocks, while low-productivity firms exit and new local firms enter.

[Table 6 here]

5.2 Decompositions by firm characteristics

Table 7 decomposes the estimated average impact of Russian immigration over the post-war period (2022–2023) on employment growth by firm category (size–age), splitting the effects into firm dynamics and ownership components. It is important to note that the decomposition is normalized, within each breakdown (by firm dynamics or by ownership), the reported coefficients sum to one. For example, in the "firm dynamics" decomposition, the contributions of continuers, entrants, and exiters add up to the total within each size–age group (e.g., $0.181 + 0.187 - 0.009 \approx 0.358$ for small–young firms). Hence, each coefficient can be interpreted as the share of total employment growth attributable to that component, while the total coefficient reflects the overall magnitude of the immigration shock on employment growth for that firm category.

[Table 7 here]

Column (1) reports the total effect, columns (2)–(4) break this down by firm dynamics (continuers, entry, and exit), and columns (5)–(6) decompose the total into domestic- and foreign-owned firms. The results show that the positive aggregate employment effects are driven almost entirely by small–young and small–old firms. For small–young firms, the total impact is 0.358, with contributions split almost evenly between continuers (0.181) and entry (0.187). For small–old firms, the total effect is also positive (0.244), but the composition differs: the main source of growth comes from continuers (0.346), while exit contributes negatively (–0.102), pointing to a process of creative destruction where the exit of older, less productive incumbents is outweighed by expansion among survivors.

Medium-sized firms also show positive total effects (0.211), driven mainly by continuing firms (0.242), with negligible contributions from entry and a small negative from exit. Large firms exhibit a modest total effect (0.187), entirely from continuing firms. Looking across ownership types, both domestic and foreign-owned firms contribute to the positive aggregate effect for small–young and small–old firms, but the pattern differs. For small–young firms, domestic firms account for most of the increase (0.239), though foreign-owned firms also grow (0.119). For small–old firms, the effect is concentrated among domestic firms (0.246), with no significant contribution from foreign-owned firms.

Table 8 applies the same decomposition to firm count growth. The results show that the post-war increase in firm counts is driven almost entirely by small–young firms. Their

total growth effect (1.409) stems from new firm entry (1.360), with a modest additional contribution from reduced net exit (0.049). For clarity, the contribution of firm exits is entered with the opposite sign, so that reductions in exit rates appear as positive contributions to total growth. Thus, a positive exit coefficient indicates that fewer firms closed during the period, while a negative coefficient implies an increase in firm closures. On the ownership side, most of this expansion is among domestic firms (1.271), although small firms owned by foreign investors also contribute significantly (0.139). In contrast, small-old firms experienced a net contraction (-0.409), almost entirely explained by net exit (–0.417). This decline is concentrated among domestic firms (–0.364), with foreign-owned firms showing no significant change. Medium and large firms exhibit negligible effects, with no meaningful contributions from either firm dynamics or ownership.

[Table 8 here]

Overall, the decomposition indicates that Russian immigration during 2022–2023 spurred firm creation predominantly among small, young businesses, especially domestic ones, while accelerating exit among small, older domestic firms, consistent with a process of creative destruction where new entrants replace some less competitive incumbents.

5.3 Additional specifications and robustness

Two-stage least squares estimation for effects of migration indicators. For confirmation that our main reduced-form estimates capture effects on employment growth through the migration shock, we estimate various two-way fixed effect models using two-stage least squares (2SLS), detailed in Appendix A.2. Each estimate assesses how a main outcome variable is affected by a specific migration indicator, instrumented using either of our treatment exposure measures. We use two measures of migration inflows introduced in our descriptive analysis: (i) administrative statistics on Russian residence permits, and (ii) CV submissions from HeadHunter, Kazakhstan's largest online job-search platform.

Our results confirm that residence permits (Table A1) and CV submissions (Table A2) affect employment growth (Panel A of each table) and firm count growth (Panel B of each table). Effective F-statistics across all specifications indicate that instruments are sufficiently strong. Effects are positive and statistically significant across all specifications.

Results are robust regardless of whether we include or exclude Almaty and Astana, and whether we use the binary or continuous treatment exposure measure as the instrument.

Assessing identification threats and alternative mechanisms. We summarize here specifications and results from Appendix A.3 that are essential to ruling out the potential confounders that pose a threat to our identification strategy. This appendix section also rules out alternative mechanisms that could plausibly drive our results, such as export opportunities and pure demand effects of migration.

First, a key threat to our research design is that the estimated post-2022 employment response reflects economic channels related to border proximity or Russian trade integration, rather than migration, which may also relate to the pre-war Russian population share. We address this concern by augmenting the baseline difference-in-differences specification with additional interactions to control for these possibilities. We consider three controls: border distance, a binary indicator for above median border-distance, and a measure of trade integration with Russia. When we interact any one of these controls with War_t and include it in our DiD specification for employment growth, estimated coefficients are small and statistically insignificant, with no quantitatively important effect on our main coefficient of interest.

Second, we address whether employment and firm count effects may nonetheless be driven by new export opportunities related to Russian population share. To evaluate this possibility, Table A4 reports a sectoral decomposition of employment and firm-count growth into tradable and non-tradable industries. The estimated effects are concentrated almost entirely in non-tradable sectors, such as construction, retail, and local services, while tradable sectors show no significant response. This pattern suggests that the post-war adjustment operates mainly through domestic channels rather than an expansion of export-oriented production.

Third, we address the possibility that our results are driven by a pure demand-driven shock, whereas we have instead emphasized that migrants from Russia generated both a supply shock of skilled labor and a demand shock. If results were purely demand-driven with no role for skill, we would expect effects confined to consumption services (e.g., accommodation and retail trade) and no effects on high-skill intangible services (e.g., ICT and finance). We find positive and statistically significant results for both

consumption services and intangible services.

Robustness checks of specification. In Appendix A.4, we perform several robustness checks on our methodological choices. First, we verify that results are not sensitive to how employment is assigned within reported size bands. Alternative definitions using the midpoint and maximum of each band yield estimates very similar to the baseline (Tables A6–A7). Second, we re-estimate the specifications directly at the firm level, controlling for firm characteristics such as size, age, and ownership, and obtain coefficients consistent with the district-level results (Table A8). Third, we augment the regressions with additional controls, including output per worker interacted with year fixed effects, and find similarly stable estimates (Table A9). Finally, we restrict the data that we aggregate to the district level, by excluding the public firm registers data and using only administrative data covering firms with more than 100 employees (Table A10).

We then assess the robustness of the decomposition results. Tables A11–A12 apply the above-median specification for pre-war Russian shares to the decomposition of employment and firm-count growth, confirming that the positive post-war effects persist even under a binary exposure measure. Tables A13–A14 repeat the decompositions excluding Almaty and Astana, showing that the aggregate patterns are not driven by these two large urban centers.

Placebo specifications. In Appendix A.5, we address the concern that our results may be driven by unobserved economic shocks that coincide with the war and relate to a district's general appeal as a migration destination. We estimate a placebo specification in which we replace the Russian population share in our research design with the share of non-Russian minorities within the district's non-Russian population (Table A15). We find no statistically significant effects, supporting our claim that the 2018 Russian population share isolates factors specific to the recent Russian migration rather than broader confounding trends.

6 Conclusion

This paper provides unique evidence of how a large-scale migration shock reshapes firm dynamics in a middle-income economy. By leveraging the 2022 Russian migration shock to

Kazakhstan, we move beyond aggregate employment effects to uncover a distinct pattern of structural adjustment. Contrary to the standard view where dynamism is the exclusive preserve of young entrants, we show that small mature firms are a primary margin of adjustment. This group drove nearly a quarter of total employment growth, yet simultaneously accounted for the majority of exits. This high degree of churn suggests the shock triggered an intense selection mechanism within the incumbent population: it forced the exit of stagnant firms while loosening labor constraints for productive survivors to expand. For policymakers, this implies that the disruption caused by migration—including the visible exit of older firms—should be viewed as a necessary phase of structural modernization that reallocates resources toward the most capable domestic firms. To the extent that migrants eventually return, the employment gains from surviving and entering firms may nonetheless persist, as the reallocation of resources toward more productive incumbents represents a durable structural change rather than a purely temporary adjustment.

We highlight that these results occur in a context where Kazakhstan has leveraged e-government to reduce frictions in migration and formal employment—features not shared by all middle-income economies. A typical developing economy often features a large, low-productivity informal sector (Hsieh and Olken 2014; Tybout 2014) that absorbs immigrant labor, though positive shocks can shift this employment toward the formal sector (McCaig and Pavcnik 2018). While digitalization of state services can promote small firm growth and reduce informality (EBRD 2021; World Bank 2023; Añón Higón and Bonvin 2024), high fixed costs for hiring migrants remain a barrier even in high-income economies like Germany (Brinatti and Morales 2021). A healthy economy should respond to a migration shock with both firm entry and the faster expansion of surviving small firms (Aghion 2017); our results demonstrate that with the right institutional frictions removed, this ideal is achievable.

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Table 1: District summary statistics

Variable	Obs	Mean	S.D.	Min	Max
Russian population share for 2018 ($Share_d$)	1290	0.200	0.193	0	0.816
District above-median Russian population share 2018, median = 0.133 ($Above\text{-}median_d$)	1290	0.512	0.500	0	1
Non-Russian minority share for 2018 ($Minority_d$)	1290	0.113	0.113	0.001	0.760
Employment	1290	22915	44000	246	259008
Permanent residence permits issued to Russian citizens	1290	24	78	0	971
Number of CVs of Russian citizens	1290	58	193	0	1665
Δ log number of permanent residence permits issued to Russian citizens (ΔI)	1075	0.055	0.937	-3.497	4.189
Δ log number of CVs of Russian citizens (ΔCV)	1075	0.116	0.791	-3.258	3.689
Log output per worker	1075	12.491	1.531	6.645	17.939
Employment growth	1075	0.012	0.055	-0.457	0.395
Firm count growth	1075	0.049	0.071	-0.666	0.387

Notes: The mean and median for both the Russian population share and the non-Russian minority share are computed using unweighted averages.

Table 2: Employment growth and firm counts by size and year

Panel A: Employment growth by size and year

Year	Small young	Small old	Medium	Large
2019	0.011	0.067	0.026	0.002
2020	-0.007	0.043	0.037	-0.043
2021	0.067	0.045	-0.023	0.001
2022	0.052	0.081	0.020	0.017
2023	-0.027	0.020	0.009	0.008

Panel B: Firm counts by size and year

Year	Total number of firms	Small young	Small old	Medium	Large
2018	279055	0.502	0.435	0.053	0.008
2019	307346	0.497	0.445	0.050	0.008
2020	329464	0.488	0.456	0.049	0.007
2021	352869	0.492	0.458	0.044	0.006
2022	397977	0.490	0.464	0.040	0.006
2023	419953	0.481	0.475	0.039	0.005

Panel C: Employment share by size and year

Year	Total employment	Small young	Small old	Medium	Large
2018	5014230	0.153	0.200	0.310	0.337
2019	5136686	0.151	0.209	0.311	0.329
2020	5162591	0.149	0.217	0.321	0.313
2021	5241897	0.157	0.224	0.309	0.309
2022	5440339	0.159	0.233	0.304	0.303
2023	5544438	0.158	0.238	0.304	0.299

Notes: Size categories: Small [1–50], Medium [51–250], Large [>250]. Employment growth is computed using the average of current- and previous-year employment as the denominator (Davis et al. 1996) and weighted by average employment.

Table 3: DiD estimates for employment growth

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.064***		0.042**	
	(0.024)		(0.020)	
$Above\text{-}median_d * War_t$		0.038***		0.029***
		(0.007)		(0.006)
Districts FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.303	0.339	0.247	0.272
Districts	215	215	204	204
Observations	1075	1075	1020	1020

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table 4: DiD estimates for firm count growth

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.099***		0.060**	
	(0.032)		(0.024)	
$Above\text{-}median_d * War_t$		0.045***		0.026**
		(0.013)		(0.012)
Districts FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.520	0.536	0.522	0.526
Districts	215	215	204	204
Observations	1075	1075	1020	1020

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table 5: DiD estimates for entry rate, exit rate, and output per worker

	Entry rate (1)	Exit rate (2)	Output per worker (3)
$Share_d \times War_t$	0.035***	0.026***	0.202**
	(0.008)	(0.010)	(0.092)
R-Squared	0.787	0.427	0.943
Districts	215	215	215
Observations	1075	1075	1075

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table 6: DiD estimates for heterogeneous effects by baseline district productivity

	Entry rate (1)	Exit rate (2)	Employment growth (3)
$Share_d \times War_t$	0.037***	0.021	0.013
	(0.013)	(0.015)	(0.029)
$Share_d \times War_t \times$ Low (p0–p25)	0.075**	0.088**	0.064
	(0.032)	(0.037)	(0.072)
$Share_d \times War_t \times$ High (p75–p100)	-0.013	-0.069***	0.115***
	(0.017)	(0.020)	(0.039)
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Year × Productivity-group FE	Yes	Yes	Yes
Observations	1075	1075	1075
Districts	215	215	215

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted. The omitted productivity category is (p25–p75).

Table 7: Contribution to employment growth

	Total	By firm dynamics			By ownership	
		Continuer	Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)	(6)
Small young	0.358***	0.181**	0.187**	-0.009	0.239*	0.119**
	(0.132)	(0.077)	(0.087)	(0.049)	(0.139)	(0.057)
Small old	0.244***	0.346***	-0.001	-0.102**	0.246***	-0.002
	(0.062)	(0.071)	(0.001)	(0.045)	(0.060)	(0.015)
Medium	0.211*	0.242**	0.007	-0.039	0.246**	-0.035
	(0.113)	(0.117)	(0.022)	(0.051)	(0.102)	(0.040)
Large	0.187	0.187	0.000	0.000	0.011	0.176
	(0.186)	(0.180)	(0.000)	(0.000)	(0.211)	(0.137)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Districts	215	215	215	215	215	215
Observations	1075	1075	1075	1075	1075	1075

Notes: The table reports the decomposition of employment growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

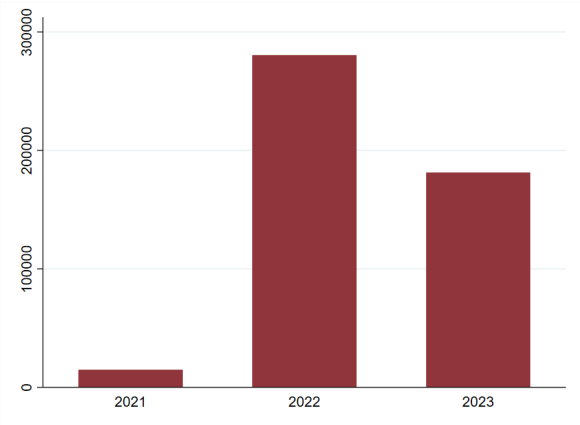
Table 8: Contribution to firm count growth

	Total	By firm dynamics		By ownership	
		Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)
Small young	1.409***	1.360***	0.049	1.271***	0.139**
	(0.150)	(0.073)	(0.050)	(0.191)	(0.064)
Small old	-0.409**	0.008	-0.417**	-0.364*	-0.045
	(0.201)	(0.022)	(0.190)	(0.202)	(0.043)
Medium	-0.001	-0.000	-0.000	-0.001	-0.000
	(0.000)	(0.009)	(0.000)	(0.004)	(0.000)
Large	0.000	0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Districts	215	215	215	215	215
Observations	1075	1075	1075	1075	1075

Notes: The table reports the decomposition of firm count growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. The exit component is reported with the opposite sign so that reductions in exit rates appear as positive contributions. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

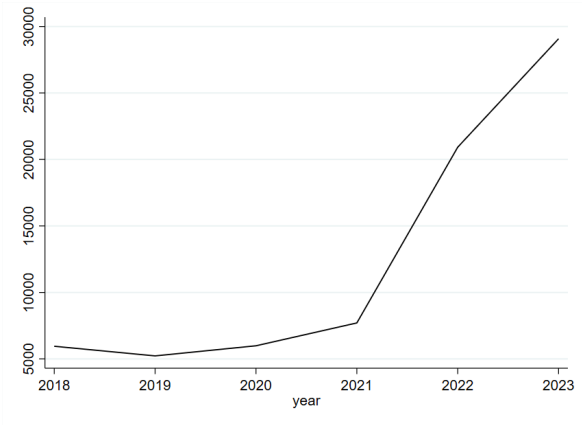
Figure 1: Migration indicators: Individual Identification Numbers (IINs) and CVs of Russian citizens

Panel A: Number of IINs issued to Russian citizens in Kazakhstan, 2021–2023



Source: Committee of the Migration Service, Ministry of Internal Affairs of the Republic of Kazakhstan

Panel B: Number of CVs of Russian citizens, 2018–2023



Source: HeadHunter (hh.kz)

Figure 2: Russian population shares by district in Kazakhstan for 2018

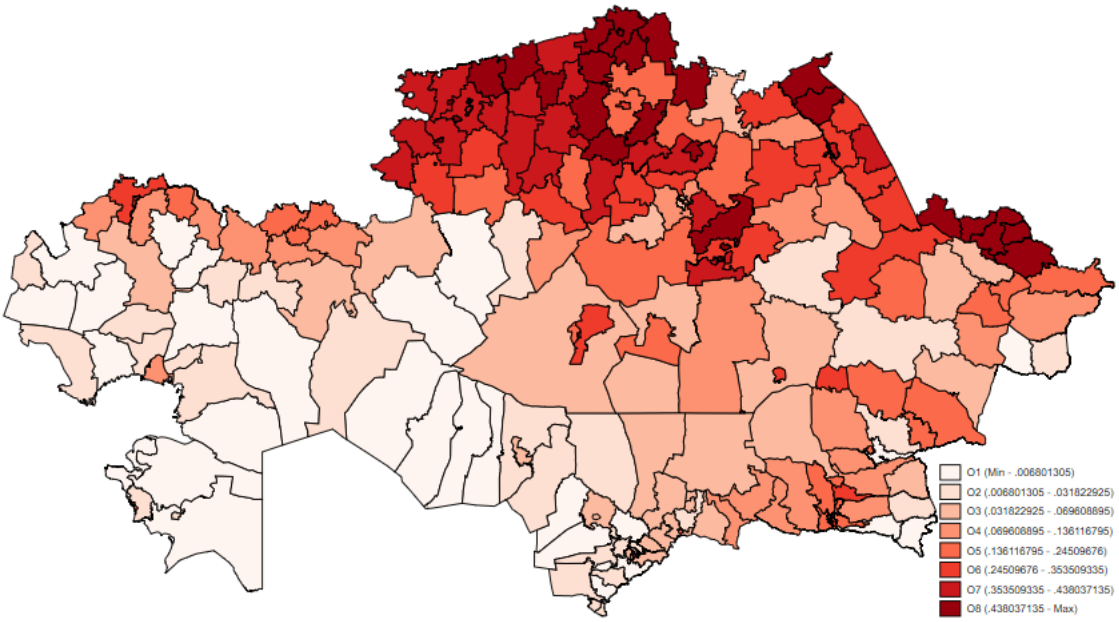
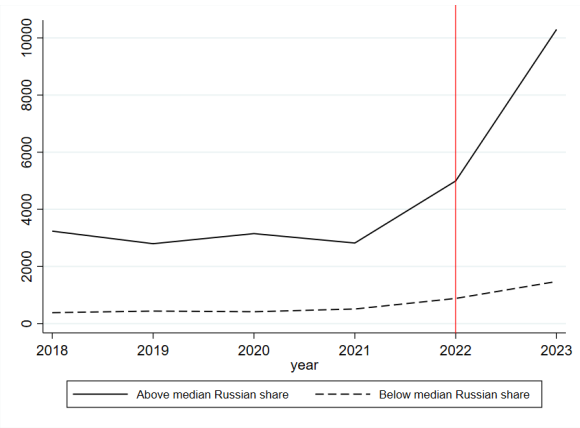


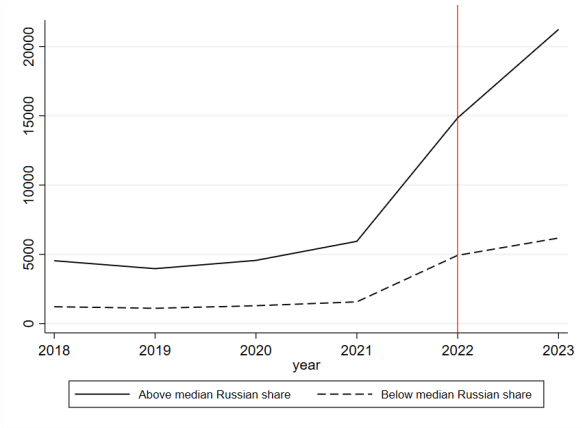
Figure 3: Migration indicators: Residence permits and CVs by Russian population share

Panel A: Number of permanent residence permits issued to Russian citizens, 2018–2023



Source: Bureau of National Statistics (BNS)

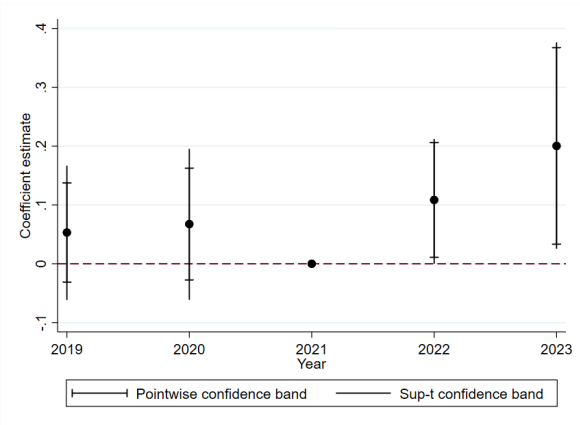
Panel B: Number of CVs of Russian citizens, 2018–2023



Source: HeadHunter (hh.kz)

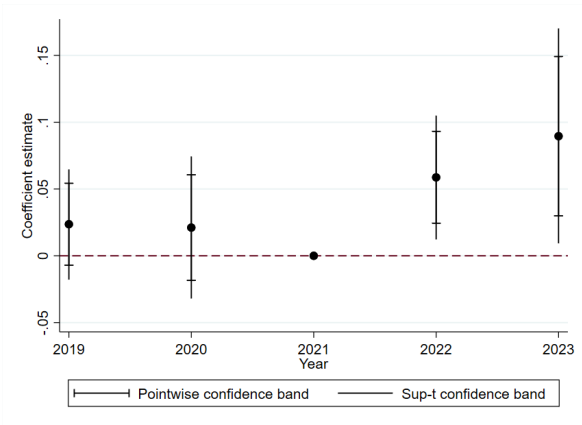
Figure 4: Event study estimation for effects on employment growth by Russian population share in 2018

Panel A: Continuous Russian share



Source: Authors' estimation using BNS data.

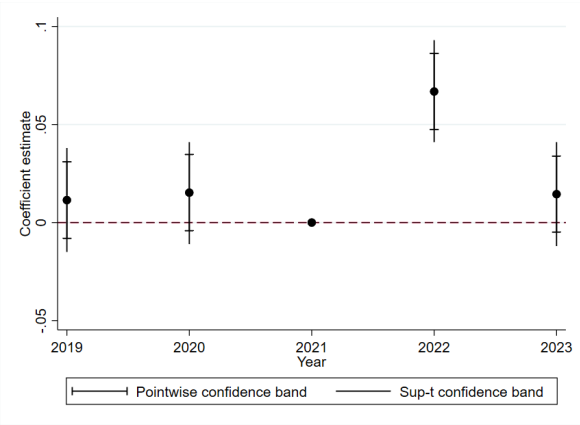
Panel B: Binary Russian share



Source: Authors' estimation using BNS data.

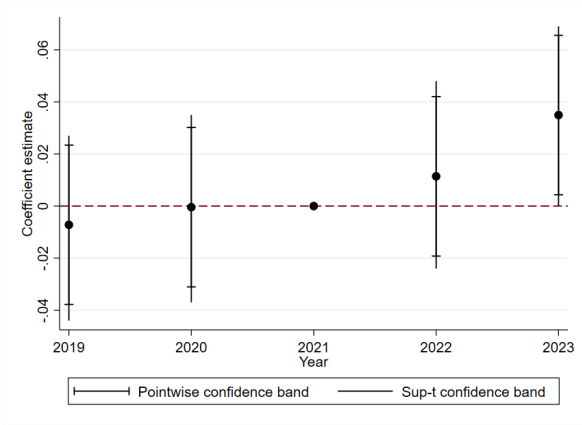
Figure 5: Event study estimation of entry rate, exit rate, and output per worker

(a) Entry rate



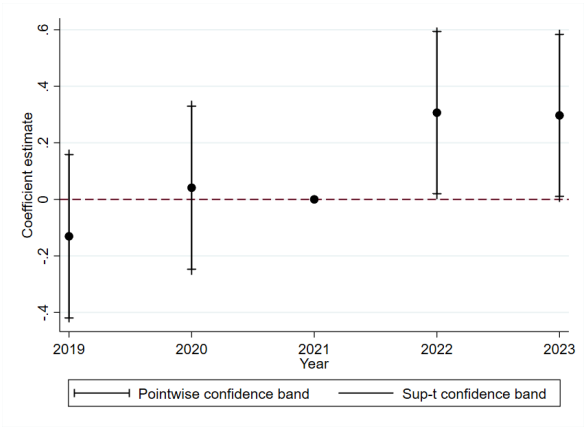
Source: Authors' estimation using BNS data.

(b) Exit rate



Source: Authors' estimation using BNS data.

(c) Output per worker



Source: Authors' estimation using BNS data.

A Appendix

A.1 Relevance of Russian population share for migration

To confirm that the 2018 pre-shock Russian population share is relevant for actual migration flows, we estimate two-way fixed effect event study specifications. The log change in a migration indicator (residence permits or CV submissions) is the dependent variable. We estimate and plot coefficients for Russian population share interacted with each year, using 2021 as the reference year.

Figure A1 reports event study estimates capturing how each migration inflow measure relates to the Russian migration share. The estimates reflect a pattern consistent with our modeling of the sudden migration shock: there are no statistically significant trends prior to 2021, while 2022 and 2023 each exhibit statistically significant increases relative to 2021.

A.2 2SLS estimation of effects of migration indicators

Because immigration may be endogenous to local firm outcomes, we implement an instrumental variables (IV) strategy based on a shift–share design. Specifically, we instrument post-war immigration inflows using the interaction of pre-war Russian population shares with the post-war period. This approach exploits plausibly exogenous variation in the geographic distribution of migration induced by the war, following established practice in the migration literature.

The first-stage regressions in Table A1, Panel A and B, show that both instruments, the continuous interaction $Share_d \times War_t$ and the above-median indicator $\textit{Above-median}_d \times War_t$, are strong, positive, and precisely estimated predictors of immigration inflows. This is evident for the specifications where the endogenous variable is the change in log residence permits issued to Russian citizens (ΔI), where the instruments are highly significant and conventional first-stage F-statistics are reported. Table A1, Panel A and B, columns (1) and (2) show that migration inflows interacted with the post-2022 war dummy are positively associated with both employment and firm count growth. For employment, a 10% increase in residence permits after the war is linked to about 0.06 percentage point faster growth, while a 10% increase in resumes corresponds to a 0.22 percentage point increase.

The effects on firms are similar in pattern: a 10% rise in permits after the war predicts 0.08 percentage point faster firm count growth, whereas resumes predict a 0.24 percentage point increase. Overall, the estimates are positive, statistically significant, and robust across both migration measures, indicating that districts receiving larger inflows following the war experienced stronger labor market and firm dynamics. Instrument strength is further supported by robust diagnostics (Table A1, Panels A and B, columns (3)–(6)). For the residence permit specifications, the Montiel–Olea and Pflueger effective F-statistics (Pflueger and Wang 2015) comfortably exceed the Stock and Yogo (Stock and Yogo 2005) critical value for a 10 percent maximal IV bias, except in column (6), where the statistic falls slightly below the threshold.

For an alternative measure of migration, we use the change in log number of resumes submitted by Russian citizens (ΔCV) as the endogenous variable (Table A2). For these specifications, we report the effective F-statistics, which again exceed the Stock and Yogo threshold. These results indicate that the instruments provide sufficiently strong and plausibly exogenous variation, supporting the credibility of the 2SLS estimates for employment and firm count growth and aligning with best practice in the IV literature (Staiger and Stock 1997; Montiel Olea and Pflueger 2013).

A.3 Checks of confounding factors and alternative explanations

A.3.1 Geographic and trade exposure

A potential concern is that the estimated post-2022 employment response reflects Russia-related trade disruption or border-driven economic channels rather than migration-induced labor market adjustment. We address this concern by augmenting the baseline difference-in-differences specification with additional controls interacted with War_t, to assess whether our main results are driven by geographic proximity to Russia and pre-war trade dependence. Border proximity controls capture the possibility that results could be driven by economic shocks that depend on closeness to Russia, such as increased trade opportunities. A pre-war trade dependence control captures the possibility that results are driven by economic shocks that relate to trade, such as increased export opportunities or import substitution.

We consider three additional controls. First, we account for geographic proximity by

interacting the post-2022 indicator with each district's log distance to the Russian border. Second, we use a binary indicator of geographic proximity based on whether each district is above or below the median border distance. Third, we construct a predetermined district-level measure of exposure to Russia trade using NACE Rev.2 2-digit industries and 2018 data. Because bilateral exports to Russia are not observed at the district level, we follow a standard shift-share approach and combine national industry-level export shares to Russia with districts' pre-war industry output composition. Let r_s denote the share of exports from industry s shipped to Russia in 2018:

$$r_s = \frac{EXP^{RU}_{s,2018}}{EXP_{s,2018}}, \tag{3}$$

where $EXP^{RU}_{s,2018}$ and $EXP_{s,2018}$ are, respectively, exports to Russia and total exports (all destinations) for industry s in 2018, aggregated at the NACE Rev.2 2-digit level.Using district-level output by industry in 2018—also measured at the NACE Rev.2 2-digit level—we compute district output shares:

$$w_{d,s} = \frac{Output_{d,s,2018}}{\sum_k Output_{d,k,2018}}, \tag{4}$$

and define district d's Russia trade exposure as the output-weighted average

$$RussiaExposure_d = \sum_s w_{d,s} \cdot r_s. \tag{5}$$

Intuitively, this index captures how strongly a district's pre-war production mix was concentrated in industries that were more export-oriented toward Russia at the national level. Because both r_s and $w_{d,s}$ are measured in 2018, the exposure measure is predetermined with respect to the 2022 geopolitical shock.

In column (3) of Table A3, we interact $RussiaExposure_d$ with the post-2022 indicator. The coefficient on the main treatment interaction ($Share_d \times War_t$) remains positive and statistically significant after conditioning on both geographic proximity and Russia trade exposure. At the same time, the additional distance- and trade-exposure interaction terms are small and statistically indistinguishable from zero.

Taken together, these results provide limited support for an interpretation centered

purely on border-driven trade disruption. Instead, the stability of the baseline estimate after conditioning on both spatial proximity and pre-war Russia trade dependence is more consistent with migration-related labor market adjustment and firm-level reallocation.

A.3.2 Mechanism check: tradable vs. non-tradable sectors

We next examine whether the post-war employment effects could be driven by export-oriented rather than domestic activities. The sectoral decomposition, presented in Table A4, separates employment and firm-count growth into tradable and non-tradable industries. The results show that the coefficients for non-tradables account for nearly all of the total effect, while those for tradables are statistically insignificant. This confirms that the aggregate findings are not driven by shifts in export-oriented sectors.

A.3.3 Mechanism check: skill-intensive vs. consumption sectors

Table A5 tests whether post-2022 employment growth in more Russia-exposed districts is concentrated in knowledge-based, skill-intensive services or in locally consumed, low-skill services, helping distinguish a local-demand response from broader structural reallocation. We classify ICT, finance, and professional/scientific activities as high-skill, intangible-intensive services, and accommodation/food, retail trade, and personal services as consumption services.

Column (1) shows a positive and statistically significant post-2022 employment response in high-skill services in more exposed districts. This pattern is consistent with complementarity mechanisms: migrant inflows—especially skilled workers—may relax human-capital constraints and support expansion in intangible-intensive activities. Column (2) shows that employment also rises significantly in consumption services, consistent with a local demand channel operating through population inflows and increased spending.

Overall, the adjustment is not confined to consumption-facing activities. The presence of significant gains in high-skill, intangible-intensive services suggests that the post-2022 response reflects more than a temporary demand boost and is consistent with a broader reallocation of activity toward sectors where human capital and knowledge inputs are central.

A.4 Robustness checks of specification

A.4.1 Specification and measurement robustness

We next confirm that our results are not sensitive to the choice of how employment is assigned within reported size bands. While our baseline specification uses the minimum point of the band, we alternatively take employment to be the midpoint and the maximum of the band and recalculate our employment growth measures. Tables A6 and A7 report these results for employment growth using the midpoint and maximum definitions, respectively. Although the coefficients show some variation relative to our baseline estimates, the overall patterns remain consistent, reinforcing that our findings are not driven by the particular band definition.

We present additional robustness checks corresponding to alternative specifications described in the main text. First, we re-estimate the specifications at the firm level without firm fixed effects, which allows us to retain new entrants in the sample while including additional controls for firm-level heterogeneity. The results, reported in Table A8, remain highly consistent with our baseline. Second, we augment the district-level regressions by adding controls for output per worker interacted with year fixed effects to capture broader local economic trends. As shown in Table A9, the estimates remain closely aligned with the main results. Finally, we restrict the sample to confidential data covering only firms with more than 100 employees, continuing to estimate at the district level. Table A10 reports these results, which remain robust and further validate the main findings.

A.4.2 Robustness of decomposition results

We next verify the robustness of the decomposition results. Tables A11–A12 apply the above-median specification for pre-war Russian shares to the decomposition of employment and firm-count growth, confirming that the positive post-war effects persist even under a binary exposure measure. Tables A13–A14 repeat the decomposition excluding Almaty and Astana, showing that the aggregate patterns are not driven by these two large urban centers.

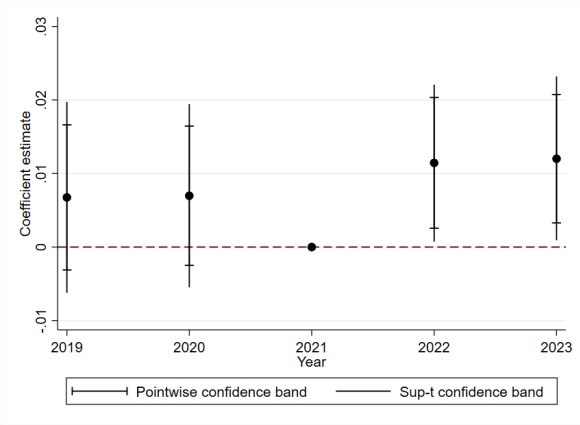
A.5 Placebo specification

Lastly, we estimate a placebo specification that substitutes for the Russian population share another population share that we expect to be unrelated to Russian immigration. The share we choose for the placebo specification is the number of non-Russian minorities divided by the non-Russian population in each district. We use all non-Russians as the denominator (this would equal non-Russian minorities plus native ethnic Kazakhs) because including Russians would lead to a share variable that is inevitably negatively correlated with the Russian share. This placebo test can help to confirm that we are capturing the effects of 2022 Russian immigration and not just effects related to the appeal of locations common for all immigrants. Our placebo share variable is summarized in Table 1. We refer to this variable as $minority_d$.

Table A15 reports the results. The placebo estimates fail to reject the hypothesis of a null effect of non-Russian minority shares for all four columns, confirming that our results specifically capture the impact of Russian migration rather than reflecting general attractiveness to all immigrant groups.

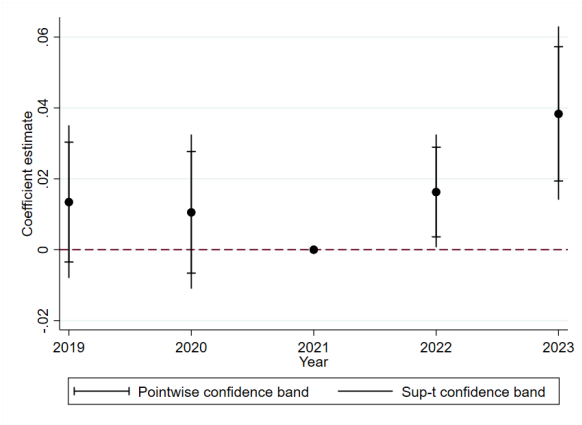
Figure A1: Event study estimation of migration inflows from Russia

Panel A: Residence permits



Source: Authors' estimation using BNS data.

Panel B: CV submissions



Source: Authors' estimation using BNS data.

Table A1: DiD and 2SLS estimates of effects of Russian residence permits (ΔI)

Panel A. Employment growth						
	$\Delta I * War_t$		ΔI Instrumented by $Share_d \times War_t$		ΔI Instrumented by $Above\text{-}median_d \times War_t$	
	(1)	(2)	(3)	(4)	(5)	(6)
	0.006***	0.005***	0.045**	0.026**	0.081***	0.052**
	(0.002)	(0.002)	(0.022)	(0.012)	(0.029)	(0.022)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Almaty–Astana incl.	Yes	No	Yes	No	Yes	No
R-squared	0.341	0.261				
Eff. F-statistic			25.628	24.207	27.129	22.029
Districts	215	204	215	204	215	204
Observations	1075	1020	1075	1020	1075	1020
Panel B. Firm count growth						
	$\Delta I * War_t$		ΔI Instrumented by $Share_d \times War_t$		ΔI Instrumented by $Above\text{-}median_d \times War_t$	
	(1)	(2)	(3)	(4)	(5)	(6)
	0.008**	0.004*	0.094***	0.041**	0.122***	0.051**
	(0.003)	(0.003)	(0.028)	(0.018)	(0.033)	(0.024)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Almaty–Astana incl.	Yes	No	Yes	No	Yes	No
R-squared	0.421	0.419				
Eff. F-statistic			28.461	24.777	29.961	17.270
Districts	215	204	215	204	215	204
Observations	1075	1020	1075	1020	1075	1020

Notes: Standard errors clustered at the district level are reported in parentheses. The *, ** and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by district employment. Columns (1)–(2) report difference-in-differences estimates using the interaction $\Delta I \times War_t$. Columns (3)–(6) report two-stage least squares (2SLS) estimates, where the endogenous variable ΔI is instrumented either with(i) the interaction between the pre-war Russian population share and the post-war period ($Share_d \times War_t$), or (ii) the interaction between an indicator for districts above the 2018 median Russian population share and the post-war period ($Above\text{-}median_d \times War_t$). The reported effective F-statistics correspond to the Montiel Olea–Pflueger (2013) robust weak-instrument test. The critical value of 23.109 corresponds to a maximal size distortion (τ) of 10%.

Table A2: DiD and 2SLS estimates of effects of Russian CVs (ΔCV)

Panel A. Employment growth						
	$\Delta CV * War_t$		ΔCV Instrumented by $Share_d \times War_t$		ΔCV Instrumented by $Above\text{-}median_d \times War_t$	
	(1)	(2)	(3)	(4)	(5)	(6)
	0.022***	0.020***	0.108***	0.052***	0.147***	0.074***
	(0.004)	(0.003)	(0.042)	(0.020)	(0.043)	(0.018)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Almaty–Astana incl.	Yes	No	Yes	No	Yes	No
R-squared	0.370	0.299				
Eff. F-statistic			25.777	25.307	26.764	30.939
Districts	215	204	215	204	215	204
Observations	1075	1020	1075	1020	1075	1020
Panel B. Firm count growth						
	$\Delta CV * War_t$		ΔCV Instrumented by $Share_d \times War_t$		ΔCV Instrumented by $Above\text{-}median_d \times War_t$	
	(1)	(2)	(3)	(4)	(5)	(6)
	0.024***	0.013***	0.168***	0.056**	0.175***	0.044***
	(0.006)	(0.005)	(0.061)	(0.024)	(0.051)	(0.017)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Almaty–Astana incl.	Yes	No	Yes	No	Yes	No
R-squared	0.429	0.423				
Eff. F-statistic			25.316	28.326	29.200	36.135
Districts	215	204	215	204	215	204
Observations	1075	1020	1075	1020	1075	1020

Notes: Standard errors clustered at the district level are reported in parentheses. The *, ** and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively. All regressions are weighted by district employment. Columns (1)–(2) report difference-in-differences estimates using the interaction $\Delta CV \times War_t$. Columns (3)–(6) report two-stage least squares (2SLS) estimates, where the endogenous variable ΔCV is instrumented either with (i) the interaction between the pre-war Russian population share and the post-war period ($Share_d \times War_t$), or (ii) the interaction between an indicator for districts above the 2018 median Russian population share and the post-war period ($Above\text{-}median_d \times War_t$). The reported effective F-statistics correspond to the Montiel Olea–Pflueger (2013) robust weak-instrument test. The critical value of 23.109 corresponds to a maximal size distortion (τ) of 10%.

Table A3: Controlling for Geographic and Trade Exposure to Russia

	Distance Interaction (1)	Close-to-Russia Interaction (2)	Exposure Interaction (3)
$Share_d \times War_t$	0.065***	0.062***	0.060***
	(0.023)	(0.024)	(0.023)
$Distance_d \times War_t$	-0.003		
	(0.002)		
$Close_d \times War_t$		0.001	
		(0.008)	
$Exposure_d \times War_t$			-0.015
			(0.022)
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	1075	1075	1075
Districts	215	215	215

Notes: Standard errors are clustered by district. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively. All regressions are employment-weighted.

Table A4: Decomposition of employment effects into tradable and non-tradable sectors

	Employment growth		Firm count growth	
	Tradables	Non-tradables	Tradables	Non-tradables
	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.092	0.908***	0.025	0.975***
	(0.132)	(0.132)	(0.027)	(0.027)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Districts	215	215	215	215
Observations	1075	1075	1075	1075

Notes: The table reports the decomposition of employment and firm count growth by tradable and non-tradable sectors. Within each decomposition (employment growth and firm count growth), the coefficients across tradables and non-tradables sum to one. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A5: Sectoral heterogeneity: high-skill/intangible vs. consumption services

	High-skill/intangible services	Consumption services
	(1)	(2)
$Share_d \times War_t$	0.056***	0.123**
	(0.019)	(0.051)
District FE	Yes	Yes
Year FE	Yes	Yes
Observations	1075	1075
Districts	215	215

Notes: High-skill/intangible services include ICT, finance, professional and scientific activities. Consumption services include accommodation, food services, retail, and personal services. Standard errors clustered at the district level. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A6: DiD estimates for employment growth, using the midpoint of the range

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.064***		0.041**	
	(0.023)		(0.020)	
$Above\text{-}median_d * War_t$		0.036***		0.027***
		(0.007)		(0.006)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.330	0.363	0.252	0.275
Districts	215	215	204	204
Observations	1075	1075	1020	1020

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A7: DiD estimates for employment growth, using the maximum point of the range

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.080***		0.049**	
	(0.028)		(0.023)	
$Above\text{-}median_d * War_t$		0.041***		0.029***
		(0.008)		(0.007)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.371	0.401	0.268	0.285
Districts	215	215	204	204
Observations	1075	1075	1020	1020

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A8: DiD estimates for employment growth using firm level data

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.030***		0.033***	
	(0.011)		(0.011)	
$Above\text{-}median_d * War_t$		0.013***		0.015***
		(0.004)		(0.005)
Size FE	Yes	Yes	Yes	Yes
Sector FE	Yes	Yes	Yes	Yes
Ownership FE	Yes	Yes	Yes	Yes
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.833	0.833	0.836	0.836
Districts	215	215	204	204
Observations	2003277	2003277	1180159	1180159

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A9: DiD estimates for employment growth with additional control variables

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.071***		0.034*	
	(0.018)		(0.018)	
$Above\text{-}median_d * War_t$		0.044***		0.029***
		(0.006)		(0.007)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Output per worker - Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.294	0.324	0.226	0.239
Districts	215	215	204	204
Observations	1075	1075	1020	1020

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A10: DiD estimates for employment growth using firms with employment more than 100

	(1)	(2)	(3)	(4)
$Share_d * War_t$	0.057***		0.052***	
	(0.020)		(0.019)	
$Above\text{-}median_d * War_t$		0.042***		0.038***
		(0.009)		(0.010)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	Yes	No	No
R-Squared	0.373	0.395	0.279	0.299
Districts	206	206	195	195
Observations	1030	1030	975	975

Notes: Standard errors are clustered by districts. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A11: Contribution to employment growth (above-median specification)

	Total	By firm dynamics			By ownership	
		Continuer	Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)	(6)
Small young	0.277***	0.139*	0.161***	-0.023	0.172*	0.105***
	(0.084)	(0.077)	(0.056)	(0.031)	(0.101)	(0.035)
Small old	0.215***	0.301***	0.009	-0.095**	0.204***	0.011
	(0.039)	(0.073)	(0.011)	(0.044)	(0.038)	(0.007)
Medium	0.204***	0.201***	0.037	-0.034	0.207***	-0.003
	(0.068)	(0.075)	(0.022)	(0.028)	(0.060)	(0.019)
Large	0.305***	0.305***	0.000	0.000	0.192**	0.113*
	(0.116)	(0.121)	(0.000)	(0.000)	(0.087)	(0.065)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Districts	215	215	215	215	215	215
Observations	1075	1075	1075	1075	1075	1075

Notes: Estimates reported here use the indicator $Above\text{-}median_d$ instead of $Share_d$ in the DiD specification. The table reports the decomposition of employment growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A12: Contribution to firm count growth (above-median specification)

	Total	By firm dynamics		By ownership	
		Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)
Small young	1.356***	1.331***	0.025	1.024***	0.332***
	(0.100)	(0.131)	(0.030)	(0.214)	(0.089)
Small old	-0.356**	0.005	-0.361**	-0.321**	-0.035
	(0.167)	(0.022)	(0.181)	(0.151)	(0.043)
Medium	0.000	-0.000	0.000	0.000	0.000
	(0.000)	(0.001)	(0.000)	(0.000)	(0.000)
Large	0.000	0.000	0.000	0.000	0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Districts	215	215	215	215	215
Observations	1075	1075	1075	1075	1075

Notes: Estimates reported here use the indicator $Above\text{-}median_d$ instead of $Share_d$ in the DiD specification. The table reports the decomposition of firm count growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. The exit component is reported with the opposite sign so that reductions in exit rates appear as positive contributions. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A13: Contribution to employment growth (without Almaty and Astana)

	Total	By firm dynamics			By ownership	
		Continuer	Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)	(6)
Small young	0.286**	0.121	0.245**	-0.080	0.129*	0.156**
	(0.141)	(0.112)	(0.113)	(0.065)	(0.067)	(0.073)
Small old	0.155	0.277**	0.000	-0.122**	0.203	-0.048
	(0.129)	(0.136)	(0.000)	(0.061)	(0.128)	(0.045)
Medium	0.189	0.259	-0.027	-0.043	0.327	-0.138
	(0.245)	(0.207)	(0.041)	(0.117)	(0.252)	(0.128)
Large	0.369	0.369	0.000	0.000	-0.031	0.401
	(0.365)	(0.299)	(0.000)	(0.000)	(0.531)	(0.396)
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Districts	204	204	204	204	204	204
Observations	1020	1020	1020	1020	1020	1020

Notes: The table reports the decomposition of employment growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A14: Contribution to firm count growth (without Almaty and Astana)

	Total	By firm dynamics		By ownership	
		Entry	Exit	Domestic	Foreign
	(1)	(2)	(3)	(4)	(5)
Small young	1.142***	1.032***	0.110	0.814***	0.328**
	(0.221)	(0.130)	(0.074)	(0.309)	(0.155)
Small old	-0.142	0.012	-0.154**	-0.117**	-0.025
	(0.102)	(0.032)	(0.075)	(0.054)	(0.021)
Medium	-0.001	-0.000	-0.000	-0.001	-0.000
	(0.015)	(0.009)	(0.000)	(0.030)	(0.000)
Large	-0.000	0.000	-0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Districts	204	204	204	204	204
Observations	1020	1020	1020	1020	1020

Notes: The table reports the decomposition of firm count growth by size–age category. Within each decomposition (Total, By firm dynamics, and By ownership), coefficients across size–age groups sum to one. For "By firm dynamics" and "By ownership", the row sum across components equals the total effect for that size–age group. The exit component is reported with the opposite sign so that reductions in exit rates appear as positive contributions. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.

Table A15: Placebo DiD (non-Russian minority share specification)

	Employment growth		Firm count growth	
	(1)	(2)	(3)	(4)
$Minority_d * War_t$	-0.027	-0.017	0.050	-0.031
	(0.048)	(0.026)	(0.138)	(0.044)
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Almaty-Astana included	Yes	No	Yes	No
R-Squared	0.432	0.332	0.303	0.382
Districts	215	204	215	204
Observations	1075	1020	1075	1020

Notes: Estimates reported here use $Minority_d$ (the share of non-Russian minorities within the district's non-Russian population) instead of $Share_d$ in the DiD specification. Standard errors clustered by districts are in parentheses. The *, ** and *** indicate statistical significance at the 10%, 5% and 1% levels, respectively. All regressions are employment-weighted.